

HDOC DAY-42

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Batch-15

>QUESTION

Write a **menu-driven C program** to perform the following operations using **separate functions**:

1. Remove all vowels from a string.
2. Find the first repeating lowercase alphabet in a string.
3. Check whether two strings are anagrams of each other.

Aim

To write a menu-driven C program using separate user-defined functions to perform the following string operations:

1. Remove all vowels from a string.
2. Find the first repeating lowercase alphabet in a string.
3. Check whether two strings are anagrams.

Algorithm

Step 1: Start the program.

Step 2: Declare the required variables and strings.

Step 3: Display the menu with four choices:

- Remove vowels
- Find first repeating lowercase alphabet
- Check anagrams
- Exit

>C program;

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-:day42.c - Mousepad

1 #include <stdio.h>
2 #include <string.h>
3 void removeVowels(char s[]) {
4     int i, j = 0;
5     for(i = 0; s[i] != '\0'; i++) {
6         if(s[i] != 'a' & s[i] != 'e' & s[i] != 'i' & s[i] != 'o' & s[i] != 'u')
7             s[j++] = s[i];
8     }
9     s[j] = '\0';
10    printf("Result: %s", s);
11}
12 void repeating(char s[]) {
13    int count[26] = {0};
14    for(i = 0; s[i] != '\0'; i++)
15        count[s[i] - 'a']++;
16    for(i = 0; i < 26; i++) {
17        if(count[i] > 1) {
18            printf("First repeating: %c", i + 'a');
19            return;
20        }
21    }
22    printf("No repeating alphabet\n");
23}
24 void anagram(char a[], char b[]) {
25    int count[26] = {0};
26    for(i = 0; a[i] != '\0'; i++)
27        count[a[i] - 'a']++;
28    for(i = 0; b[i] != '\0'; i++)
29        count[b[i] - 'a']--;
30    for(i = 0; i < 26; i++) {
31        if(count[i] != 0) {
32            printf("Not anagram\n");
33            return;
34        }
35    }
36    printf("Anagram\n");
37}
38 int main() {
39     char a[100], b[100];
40     int ch;
41     do {
42         printf("\n1. Remove Vowels");
43         printf("\n2. First Repeating Alphabet");
44         printf("\n3. Check Anagram");
45         printf("\n4. Exit");
46         printf("\nEnter choice: ");
47         scanf("%d", &ch);
48         if(ch > 4 || ch < 1) {
49             printf("Enter string: ");
50             scanf("%s", a);
51             printf("Result: %s", a);
52             break;
53         }
54         case 1: removeVowels(a); break;
55         case 2: repeating(a); break;
56         case 3: printf("Enter second string: ");
57             scanf("%s", b);
58             anagram(a, b);
59             break;
60         case 4: printf("Exit\n"); break;
61         default: printf("Invalid choice\n");
62     }
63     while(ch != 4);
64     return 0;
65 }
```

>OUTUT

```
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali:~]$ ./day42
[priyadarshna@kali:~]$ gcc day42.c -o day42
[priyadarshna@kali:~]$ ./day42
[priyadarshna@kali:~]$ ./day42
[priyadarshna@kali:~]$ ./day42

1. Remove Vowels
2. First Repeating Alphabet
3. Check Anagram
4. Exit
Enter choice: 1
Enter string: educations
Result: dctions

1. Remove Vowels
2. First Repeating Alphabet
3. Check Anagram
4. Exit
Enter choice: 2
Enter string: apple
First repeating: p

1. Remove Vowels
2. First Repeating Alphabet
3. Check Anagram
4. Exit
Enter choice: 3
Enter string: listen
Enter second string: 4
Not Anagram

1. Remove Vowels
2. First Repeating Alphabet
3. Check Anagram
4. Exit
Enter choice: 3
Enter string: listen
Enter second string: silent
Anagram

1. Remove Vowels
2. First Repeating Alphabet
3. Check Anagram
4. Exit
Enter choice: 4
Exit
[priyadarshna@kali:~]$
```